

Michael Schaaf

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EDUCATION

SUNY Cortland Cortland, New York

Bachelor's Degree in Business Economics August 2014 - May 2017

Iona College New Rochelle, New York

Associate's in Liberal Arts August 2012 - May 2014

PROJECTS

Solo Travel Companion App

- Designed and built a full-stack travel app for solo travelers to discover, book, and safely experience activities in any city — with a React/Tailwind frontend and a FastAPI/PostgreSQL/PostGIS backend deployed with Docker.
- Implemented a QR-based buddy system allowing users to share booked reservations with nearby travelers via scannable invite codes with phone-verified guest join flows.
- Built a Safety Guardian module with scheduled check-ins, missed-check-in escalation to trusted contacts via Twilio SMS, and a one-tap panic mode that broadcasts GPS location to emergency contacts.
- Integrated Claude API to generate personalized AI itineraries from real activity data, with a mood toggle (chill vs. adventure) that adapts recommendations to the traveler's energy level.
- Architected real-time Social Radar proximity detection using PostGIS spatial queries and WebSockets, enabling opt-in discovery of nearby solo travelers with privacy controls.

Echo - Voice-First Language Translation & Learning App

- Built a voice-first iOS translation and language learning app using React Native for the frontend and a FastAPI/PostgreSQL backend deployed with Docker.
- Integrated Claude API for context-aware translation, OpenAI Whisper for speech-to-text transcription, and ElevenLabs/gTTS for natural text-to-speech output.
- Designed a freemium monetization model with tiered usage caps, optimizing per-request cost to ~\$0.02 by routing free-tier users through gTTS and premium users through ElevenLabs.
- Implemented user authentication, session history, and language preference storage with SQLAlchemy ORM and Pydantic validation.

DocuMind - RAG Document Q&A System

- Built a retrieval-augmented generation application enabling natural language Q&A over uploaded documents using Python, LangChain, and FAISS for vector similarity search.
- Implemented document chunking, OpenAI embedding generation, and LLM-powered response synthesis with source citation grounding.
- Designed a FastAPI service layer with file upload handling, async retrieval pipelines, and structured JSON responses.

Twitch Stream Translator - Real-Time Translation Chrome Extension

- Developed a Chrome Extension that captures live Twitch audio streams and provides real-time translated captions using Deepgram and OpenAI Whisper for speech-to-text.
- Built with JavaScript (content scripts, background service worker) and integrated WebSocket-based streaming for low-latency transcription.

SKILLS

Cloud/DevOps: Azure (VMs, Resource Groups, AD, Key Vault), AWS (EC2, IAM, Secrets Manager), CI/CD pipelines (GitHub Actions), infrastructure automation, Docker, Terraform, Kubernetes

Engineering: Python, React/React Native, FastAPI, SQLAlchemy, Tailwind CSS, PowerShell, JavaScript/TypeScript, PostgreSQL, PostGIS, Redis, Celery, WebSockets

AI/ML: LangChain, RAG (embeddings + FAISS vector retrieval), MCP server (Python) for LLM tool integration, agent/tool-calling workflows, eval harnesses, Claude API, OpenAI API, Whisper STT, ElevenLabs/gTTS, Deepgram

Platforms & Services: Azure AD/Active Directory, GitHub, Intune (endpoint & policy management), Microsoft Defender, Exchange Online, SharePoint, KnowBe4, Twilio, Stripe/RevenueCat, Veeam & Wasabi, Nasuni

EXPERIENCE

Systems Engineer | PharmalytIQ 2025 - Present

- Led development of internal full-stack tools with React and Python, delivering license tracking, user lifecycle automation, and operational dashboards across four pharmacy locations.
- Managed AWS and Azure infrastructure: provisioning, IAM/policy management, secrets rotation (Key Vault / Secrets Manager), networking, and cost optimization.
- Built CI/CD pipelines and deployment automation with GitHub Actions, integrating environment promotion to reduce manual release effort.
- Collaborated on a custom Active Directory API exposing identity and group data, consumed by internal tooling and automation workflows.
- Automated operational workflows with Python and PowerShell — user provisioning/deprovisioning, permission reconciliation, audit exports, health checks — and built observability dashboards and alerting for incident response.
- Designed Power Automate flows to track license and credential expirations across all locations, eliminating missed renewals.

Systems Administrator | Dental Care Alliance 2023 - 2025

- Managed Active Directory for 5,000+ users, scripting account lifecycle automation to reduce manual errors during onboarding/offboarding.
- Ran Veeam backup infrastructure for 400+ VMs with Wasabi storage, cutting full restore time by ~30% and validating recoverability with regular checks.
- Administered Microsoft 365, Intune MDM, and Azure VMs/networks — maintaining compliance, group policies, and connectivity across the organization.
- Standardized domain joining, container organization, and SharePoint access controls to improve security posture and collaboration workflows.

Technical Support Representative | Dental Care Alliance November 2021 - 2023

- Led deployment of battery backup systems across ~400 locations, coordinating logistics, testing, and technician handoffs to maintain business continuity.
- Provided first-level Windows support and triaged hardware/software issues; documented recurring error patterns to improve stability of critical dental applications.

CERTIFICATIONS

Microsoft AZ-104 CompTIA Security+ CompTIA Network+